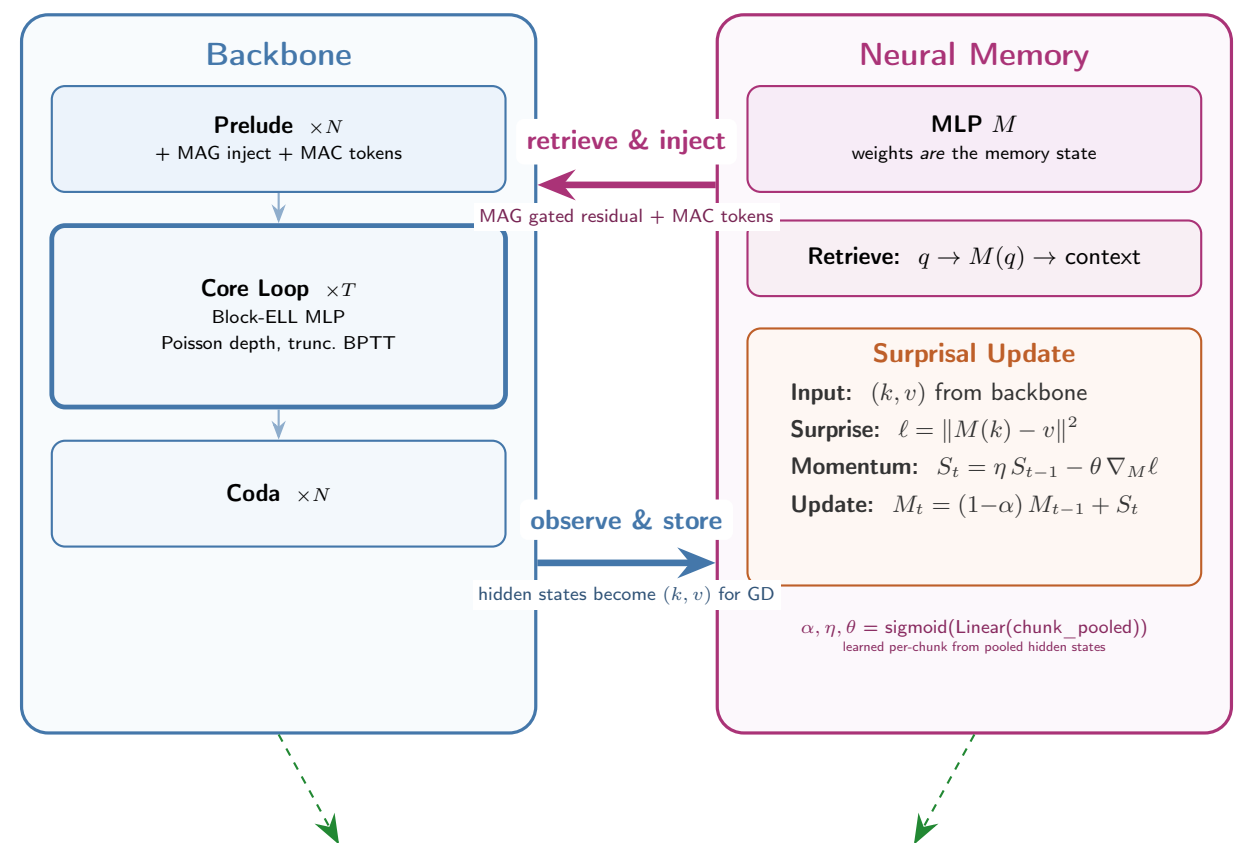


Neural Memory & Z-Latent Prediction



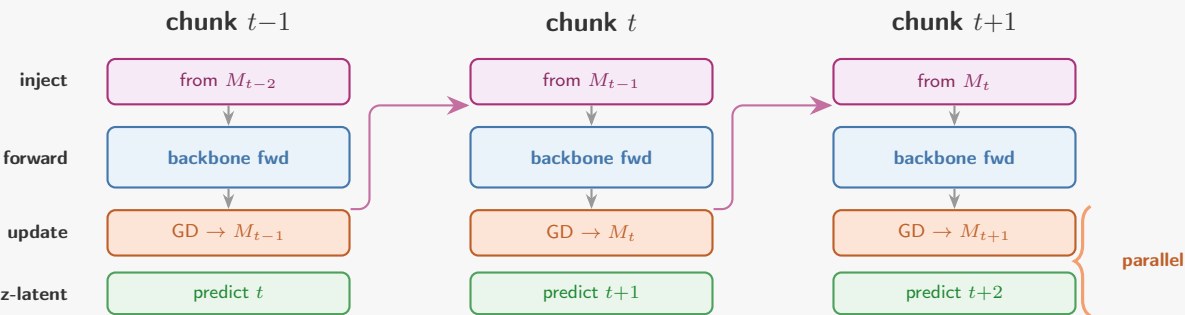
Z-Latent Objective

Trains memory **what to remember** by predicting the future
Backbone head $\rightarrow$ next window's coda repr. Memory head $\rightarrow$ next window's prelude state

How Hidden States Become (k, v) for Gradient Descent

- Backbone processes a **chunk** of tokens through prelude $\rightarrow$ core $\rightarrow$ coda
- Hidden states h are projected: $k = W_k(h)$ (address), $v = W_v(h)$ (content to store)
- Memory queries itself: $M(k) = \text{"what do I already know at this address?"}$
- Surprise** $= \|M(k) - v\|^2$ — high if v is new, low if already stored
- Gradient descent on M 's weights: surprised $\rightarrow$ large update, unsurprised $\rightarrow$ small update

Timeline: Operations Across Chunks



M = neural memory MLP (weights are the stored state)

$\downarrow$ = sequential $\nearrow$ = cross-chunk dependency (update at $t-1$ feeds inject at t) $\}$ = parallel (update + z-latent run together)